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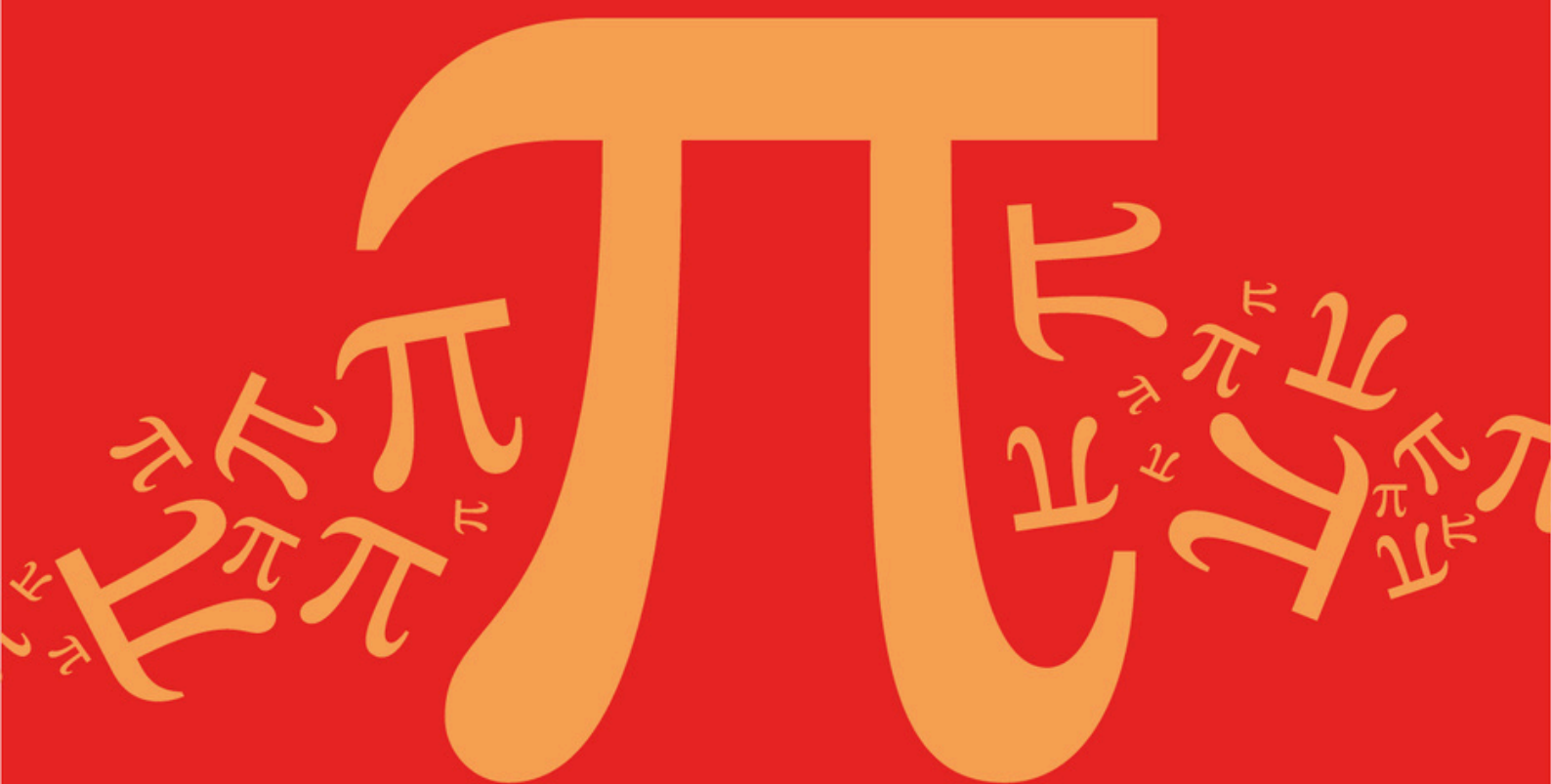
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# AQA A Level Further Mathematics (7367)



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Topic

**C: Matrices**

Sub topic

**C5: Determinants (2x2 Determinants)**

## Topic Questions

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# AQA A Level Further Mathematics (7367)

## Question Paper

Topic

**C: Matrices**

Sub topic

**C5: Determinants (2x2 Determinants)**

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for  
AQA A Level Further Mathematics (7367)  
for exams from 2025 onwards.**

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Name : \_\_\_\_\_

Date : \_\_\_\_\_

Time : \_\_\_\_\_

Total Marks Available : \_\_\_\_\_

Total Marks Received : \_\_\_\_\_

**Q1.**

- (a) Show that the matrix  $\begin{bmatrix} 5-k & 2 \\ k^3+1 & k \end{bmatrix}$  is singular when  $k = 1$ . (1)

- (b) Find the values of  $k$  for which the matrix  $\begin{bmatrix} 5-k & 2 \\ k^3+1 & k \end{bmatrix}$  has a negative determinant.  
Fully justify your answer. (5)

(Total 6 marks)

**Q2.**

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & k \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- (a) Find the value of  $k$  for which matrix  $\mathbf{A}$  is singular. (1)
- (b) Describe the transformation represented by matrix  $\mathbf{B}$ . (1)

- (c) (i) Given that  $\mathbf{A}$  and  $\mathbf{B}$  are both non-singular, verify that  $\mathbf{A}^{-1}\mathbf{B}^{-1} = (\mathbf{BA})^{-1}$ . (4)

- (ii) Prove the result  $\mathbf{M}^{-1}\mathbf{N}^{-1} = (\mathbf{NM})^{-1}$  for all non-singular matrices  $\mathbf{M}$  and  $\mathbf{N}$ . (4)

(Total 10 marks)

**Q3.**

The plane transformation  $T$  is defined by

$$T : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) A shape has an area of 3 square units. Find the area of the shape after being transformed by  $T$ . (2)

- (b) (i) Find the equations of all the invariant lines of  $T$ . (5)

- (ii) State the equation of the line of invariant points of  $T$ . (1)

(Total 8 marks)

**Q4.**

- (a) Find the eigenvalues and corresponding eigenvectors of the matrix

$$\mathbf{M} = \begin{bmatrix} 4 & 5 \\ 5 & 4 \end{bmatrix}.$$

(6)

- (b) The plane transformation  $T$  is given by the matrix  $\mathbf{M}$ . Write down the coordinates of the invariant point of  $T$ .

(1)

(Total 7 marks)

**Q5.**

Let  $\mathbf{X} = \begin{bmatrix} 3 & x \\ -1 & 7 \end{bmatrix}$ .

- (a) Determine  $\mathbf{X} \mathbf{X}^T$ .

(2)

- (b) Show that  $\text{Det}(\mathbf{X} \mathbf{X}^T - \mathbf{X}^T \mathbf{X}) \leq 0$  for all real values of  $x$ .

(4)

- (c) Find the value of  $x$  for which the matrix  $(\mathbf{X} \mathbf{X}^T - \mathbf{X}^T \mathbf{X})$  is singular.

(1)

(Total 7 marks)

**Q6.**

The plane transformation  $T$  is a rotation through  $\theta$  radians anticlockwise about  $O$ , and maps points  $(x, y)$  onto image points  $(X, Y)$  such that

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$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

where  $c = \cos \theta$  and  $s = \sin \theta$ .

- (a) Write down the inverse of the matrix  $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}$  and hence show that

$$x = cX + sY \quad \text{and} \quad y = -sX + cY$$

(3)

- (b) The curve  $C$  has equation  $x^2 - 6xy - 7y^2 = 8$ .

The image of  $C$  under  $T$  is the curve  $C'$  with equation  $pX^2 + qXY + rY^2 = 8$ .

- (i) Use the results of part (a) to show that

$$q = 6s^2 + 16sc - 6c^2$$

and express  $p$  and  $r$  similarly in terms of  $c$  and  $s$ .

(4)

- (ii) Given that  $\theta$  is an acute angle, find the values of  $c$  and  $s$  for which  $q = 0$  and hence in this case express the equation of  $C'$  in the form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

(8)

- (iii) Hence explain why  $C$  is a hyperbola.

(1)

(Total 16 marks)

**Q7.**

The plane transformation  $T$  is represented by the matrix  $\mathbf{M} = \begin{bmatrix} -3 & 8 \\ -1 & 3 \end{bmatrix}$ .

- (a) The quadrilateral  $ABCD$  has image  $A'B'C'D'$  under  $T$ .

Evaluate  $\det \mathbf{M}$  and describe the geometrical significance of both its sign and its magnitude in relation to  $ABCD$  and  $A'B'C'D'$ .

(3)

- (b) The line  $y = px$  is a line of invariant points of  $T$ , and the line  $y = qx$  is an invariant line of  $T$ .

Show that  $p = \frac{1}{2}$  and determine the value of  $q$ .

(5)

- (c) Find the  $2 \times 2$  matrix  $\mathbf{R}$  which represents a reflection in the line  $y = \frac{1}{2}x$ .

(2)

(Total 10 marks)

**Q8.**

The  $2 \times 2$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  are such that

$$\mathbf{AB} = \begin{bmatrix} 9 & 1 \\ 7 & 13 \end{bmatrix} \quad \text{and} \quad \mathbf{BA} = \begin{bmatrix} 14 & 2 \\ 1 & 8 \end{bmatrix}$$

Without finding  $\mathbf{A}$  and  $\mathbf{B}$ :

- (a) find the value of  $\det \mathbf{B}$ , given that  $\det \mathbf{A} = 10$ ;

(3)

- (b) determine the  $2 \times 2$  matrices  $\mathbf{C}$  and  $\mathbf{D}$  given by

$$\mathbf{C} = (\mathbf{B}^T \mathbf{A}^T) \quad \text{and} \quad \mathbf{D} = (\mathbf{A}^T \mathbf{B}^T)^T$$

where  $\mathbf{M}^T$  denotes the transpose of matrix  $\mathbf{M}$ .

(3)

(Total 6 marks)

**Q9.**

Let  $\mathbf{P} = \begin{bmatrix} 1 & 4 & 2 \\ -1 & 2 & 6 \end{bmatrix}$  and  $\mathbf{Q} = \begin{bmatrix} k & 1 \\ 2 & -1 \\ 3 & 1 \end{bmatrix}$ , where  $k$  is a constant.

- (a) Determine the product matrix  $\mathbf{PQ}$ , giving its elements in terms of  $k$  where appropriate.

(3)

- (b) Find the value of  $k$  for which  $\mathbf{PQ}$  is singular.

(2)

(Total 5 marks)

**Q10.**

The plane transformation  $T$  is defined by

$$T: \begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{M} \begin{bmatrix} x \\ y \end{bmatrix}$$

where  $\mathbf{M} = \begin{bmatrix} -1 & 4 \\ -1 & 3 \end{bmatrix}$ .

- (a) Evaluate  $\det \mathbf{M}$  and state the significance of this answer in relation to  $T$ .

(2)

- (b) Find the single eigenvalue of  $\mathbf{M}$  and a corresponding eigenvector. Describe the geometrical significance of these answers in relation to  $T$ .

(5)

- (c) Show that the image of the line  $y = \frac{1}{2}x + k$  under  $T$  is  $y' = \frac{1}{2}x' + k$ .

(3)

(Total 10 marks)

**Q11.**

The transformation  $T$  maps  $(x, y)$  to  $(x', y')$  where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) Describe the difference between an *invariant line* and a *line of invariant points* of  $T$ .

(1)

- (b) Evaluate the determinant of the matrix  $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$  and describe the geometrical

significance of the result in relation to T.

(2)

(c) Show that T has a line of invariant points, and find a cartesian equation for this line.

(2)

(d) (i) Find the image of the point  $(x, -x + c)$  under T.

(2)

(ii) Hence show that all lines of the form  $y = -x + c$ , where  $c$  is an arbitrary constant, are invariant lines of T.

(2)

(e) Describe the transformation T geometrically.

(3)

(Total 12 marks)

## Q12.

The transformation T is a stretch in a fixed direction, and maps the point  $(x, y)$  to the image point  $(x', y')$ , where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(a) Determine the scale factor of the stretch.

(2)

(b) By considering the invariant points of T, or otherwise, find an equation for the line of invariant points of T, giving your answer in the form  $y = mx$ .

(5)

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(Total 7 marks)